

AMENDMENT NO. 1 JUNE 2018
TO
IS 2314 : 1986 SPECIFICATION FOR STEEL SHEET PILING SECTION
(First Revision)

(Page 3, clause 1.1) — Substitute the following for the existing:

‘1.1 This standard stipulates dimensions and dimensional tolerances for Z-type, U-type and flat-type profile of hot rolled and cold formed steel sheet piling.’

(Page 4, clause 4.1) — Insert the following new clause:

‘4.2 Cold formed steel piling sections conforming to this specifications shall be designated with the letters ISCFPS followed by the section modulus per meter of wall in cm^3 and letter symbols Z, U and F which denote Z-type, U-type and Flat-type sections, respectively.’

(Page 5, Fig. 1 and Table 1) — Substitute the following for the existing:

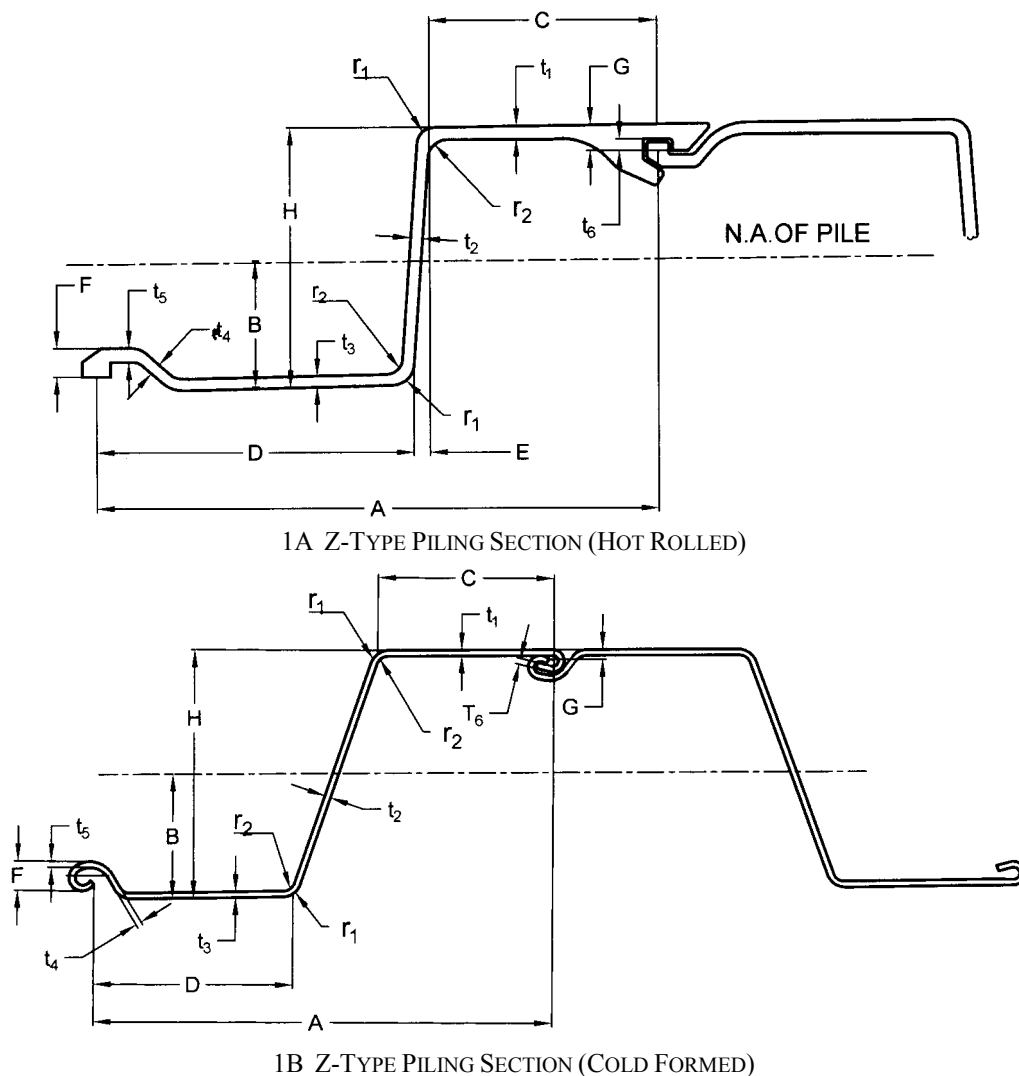


FIG. 1 Z-TYPE PILING SECTION

Price Group 5

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Table 1 Nominal Dimensions of Z-Type Piling Sections
(Clause 5.1)

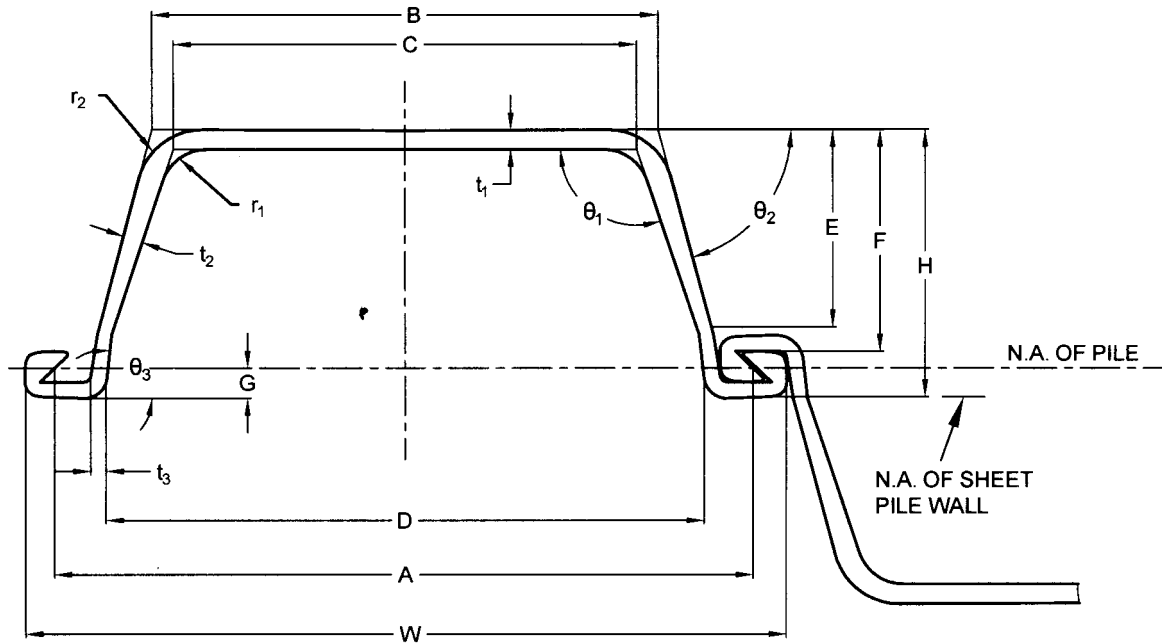
All dimensions in millimetres.

Designation	<i>H</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>t</i> ₁	<i>t</i> ₂	<i>t</i> ₃	<i>t</i> ₄	<i>t</i> ₅	<i>t</i> ₆	<i>r</i> ₁	<i>r</i> ₂
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)
a) Hot Rolled Sections																
ISPS 1021 Z	185.0	400	92.5	165.0	221.0	14.0	20.0	20.0	8.5	7.5	8.5	9.5	9.5	9.5	12.0	16.0
ISPS 1481Z	210.0	400	105.0	165.0	220.0	15.0	23.0	23.0	11.5	8.5	11.5	12.5	12.5	12.5	14.0	18.0
ISPS 1888 Z	235.0	400	117.5	165.0	218.0	17.0	24.5	24.5	13.0	9.5	13.0	14.0	14.0	14.0	15.0	20.0
ISPS 2322 Z	260.0	400	130.0	165.0	217.0	18.0	25.5	25.5	14.0	10.5	14.0	15.0	15.0	15.0	16.0	20.0
b) Cold Formed Sections																
ISCFPS 1080 Z	288	600	144	194	268	-	40	17	8	8	8	8	8	8	24	16
ISCFPS 1205 Z	303	600	151.5	217	238	-	40	17	8	8	8	8	8	8	24	16
ISCFPS 1318 Z	313	600	156.5	212	238	-	40.9	17.5	8.5	8.5	8.5	8.5	8.5	8.5	24.5	16
ISCFPS 1407 Z	326	650	163	223	271	-	41.9	18	9	9	9	9	9	9	25	16
ISCFPS 1504 Z	358	670	179	258	283	-	40	17	8	8	8	8	8	8	24	16
ISCFPS 1673 Z	400	675	200	248	253	-	40	17	8	8	8	8	8	8	24	16
ISCFPS 1759 Z	400.5	685	200.3	248	263	-	40.9	17.5	8.5	8.5	8.5	8.5	8.5	8.5	24.5	16
ISCFPS 1894 Z	404	670	202	241	283	-	41.9	18	9	9	9	9	9	9	25	16
ISCFPS 1966 Z	401.5	685	200.8	248	263	-	42.9	18.3	9.5	9.5	9.5	9.5	9.5	9.5	25.5	16
ISCFPS 2050 Z	402	685	201	248	263	-	43.9	19	10	10	10	10	10	10	26	16
ISCFPS 2117 Z	437.5	650	218.8	265	288	-	40.9	17.5	8.5	8.5	8.5	8.5	8.5	8.5	24.5	16
ISCFPS 2359 Z	445	700	222.5	258	292	-	43.9	19	10	10	10	10	10	10	26	16
ISCFPS 2500 Z	439	650	219.5	265	288	-	43.9	19	10	10	10	10	10	10	26	16
ISCFPS 2816 Z	486	675	243	292	326	-	43.9	19	10	10	10	10	10	10	26	16
ISCFPS 3046 Z	500	700	250	315	326	-	43.9	19	10	10	10	10	10	10	26	16

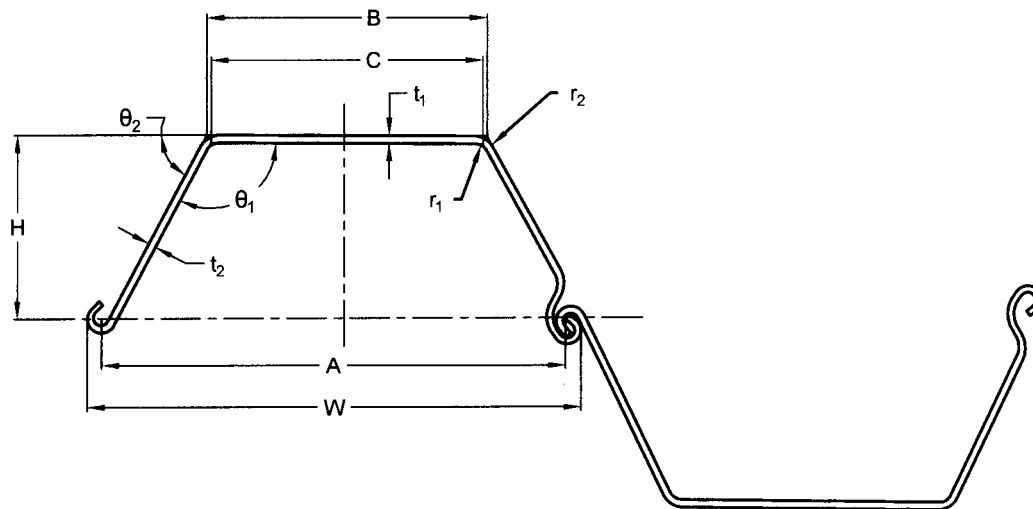
NOTE — Clutch dimension has not been specified and is left to the discretion of the manufacturer.

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(Page 6, Fig. 2 and Table 2) — Substitute the following for the existing:



2A U-TYPE HOT ROLLED



2B U-TYPE COLD FORMED

FIG. 2 U-TYPE PILING SECTION

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Table 2 Nominal Dimensions of U-Type Piling Sections
(Clause 5.2)

All dimensions in millimetres.																	
Designation	<i>W</i>	<i>H</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>t</i> ₁	<i>t</i> ₂	<i>t</i> ₃	<i>θ</i> ₁	<i>θ</i> ₂	<i>θ</i> ₃	<i>r</i> ₁	<i>r</i> ₂
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
a) Hot Rolled Sections																	
ISPS 1625 U	437	172.0	402.5	295.7	275.8	347.5	124.5	139.2	20.9	13	9.0	8.2	105	77	82.5	17.0	28
ISPS 2222 U	458	194.5	402.5	304.0	282.0	364.5	147.2	161.0	21.8	14	9.5	8.5	105	77	82.5	28.0	35
ISPS 2770 U	458	193.0	402.0	304.0	284.5	364.0	147.2	159.5	22.5	22	10	10.5	105	77	82.5	28.5	35
b) Cold Formed Sections																	
ISCFPS 986 U	639	197	600	386	377	-	-	-	-	8	8	8	120	60	-	16	24
ISCFPS 1037 U	640	197	600	386	376	-	-	-	-	8.5	8.5	8.5	120	60	-	15.5	24
ISCFPS 1100 U	641	197	600	386	376	-	-	-	-	9	9	9	120	60	-	15	24
ISCFPS 1160 U	642	197	600	386	375	-	-	-	-	9.5	9.5	9.5	120	60	-	14.5	24
ISCFPS 1450 U	643	197	600	386	375	-	-	-	-	10	10	10	120	60	-	14	24
ISCFPS 1505 U	635	260	600	424	413	-	-	-	-	8	8	8	110	70	-	16	24
ISCFPS 1593 U	636	260	600	424	412	-	-	-	-	8.5	8.5	8.5	110	70	-	15.5	24
ISCFPS 1648 U	795	233	750	342	334	-	-	-	-	9	9	9	132.2	47.8	-	15	24
ISCFPS 1687 U	637	260	600	424	411	-	-	-	-	9	9	9	110	70	-	15	24
ISCFPS 1775 U	638	260	600	424	411	-	-	-	-	9.5	9.5	9.5	110	70	-	14.5	24
ISCFPS 1860 U	639	260	600	424	410	-	-	-	-	10	10	10	110	70	-	14	24
ISCFPS 2250 U	788	240	750	451	440	-	-	-	-	10	10	10	122.2	57.8	-	14	24
ISCFPS 2581 U	682	282	650	511	497	-	-	-	-	9	9	9	105	75	-	15	24
ISCFPS 2680 U	634	250	600	478	463	-	-	-	-	10	10	10	104.8	75.2	-	14	24
ISCFPS 2815 U	617	250	600	289	278	-	-	-	-	10	10	10	122.2	57.8	-	14	24

NOTE — Clutch dimension has not been specified and is left to the discretion of the manufacturer.

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(Page 7, Fig. 3 and Table 3) – Substitute the following for the existing:

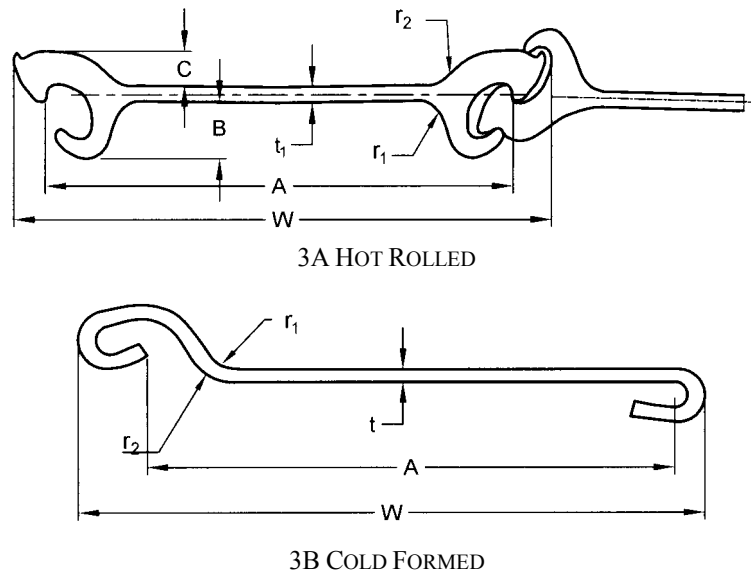


FIG. 3 FLAT-TYPE PILING SECTION

Table 3 Nominal Dimensions of Flat-Type Piling Sections
(Clause 5.3)

All dimensions in millimetres.

Designation	W	A	B	C	t	r_1	r_2
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ISPS 100 F	445.0	400.0	40.0	25.0	9.0	13.0	18.0
ISCFPS 100 F	361	300	-	-	10	16	26

NOTE — Clutch dimension has not been specified and is left to the discretion of the manufacturer.

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(Page 7, Table 4) — Substitute the following for the existing:

Table 4 Mass and Geometrical Properties of Sheet Piling Sections
(Clause 5.4)

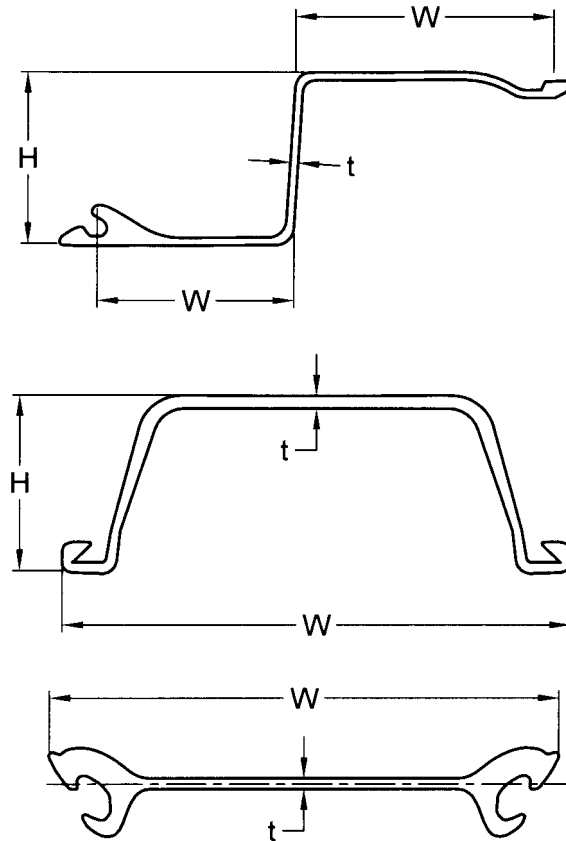
Designation	Mass Per Metre	Mass Per Square Metre of Wall	Sectional Modulus Per Metre of Wall	Moment of Inertia Per Metre of Wall	Sectional Area Per Metre of Wall	Perimeter Per Meter of Wall	Stock Width
(1)	kg (2)	kg (3)	cm ³ (4)	cm ⁴ (5)	cm ² (6)	cm (7)	mm (8)
a) Hot rolled							
ISPS 1021 Z	49.2	123.1	1 021	9 448	157	283	-
ISPS 1481 Z	63.8	159.5	1 481	15 548	203	295	-
ISPS 1888 Z	73.2	183.1	1 888	22 184	233	307	-
ISPS 2322 Z	81.6	204.1	2 322	30 189	260	318	-
ISPS 1625 Z	65.4	162.4	1 625	24 563	207	308	-
ISPS 2222 Z	82.7	195.7	2 222	38 219	249	331	-
ISPS 2770 U	95.0	226.0	2 770	47 500	288	330	-
ISPS 100 F	55.2	138.0	100	428	176	104	-
b) Cold formed							
ISCFPS 1080 Z	58.4	97.3	1 080	17 031	123.9	-	929
ISCFPS 1205 Z	58.5	97.5	1 205	18 983	124.3	-	933
ISCFPS 1318 Z	62.7	104.5	1 318	21 579	133	-	940
ISCFPS 1407 Z	70.4	108.3	1 407	24 427	149.5	-	997
ISCFPS 1504 Z	67	100	1 504	28 043	127.5	-	1 068
ISCFPS 1673 Z	68.1	100.9	1 673	34 263	128.5	-	1 084
ISCFPS 1759 Z	73.1	106.7	1 759	36 365	136	-	1 096
ISCFPS 1894 Z	77.8	116.1	1 894	40 435	148	-	1 100
ISCFPS 1966 Z	81.8	119.4	1 966	40 747	152.2	-	1 098
ISCFPS 2050 Z	85.3	124.5	2 050	41 819	158.5	-	1 086
ISCFPS 2117 Z	75.8	116.7	2 117	48 009	148.6	-	1 138
ISCFPS 2359 Z	91.56	130.8	2 359	55 041	166.6	-	1 166
ISCFPS2500 Z	89.4	137.5	2 500	56 915	175	-	1 139
ISCFPS 2816 Z	95.2	141.5	2 816	71 324	179.5	-	1 213
ISCFPS 3046 Z	100.1	143	3 046	77 902	182.1	-	1 275
ISCFPS 986 U	61.28	102.13	986	22 630	130	-	976
ISCFPS1037 U	65.13	108.55	1 037	23 967	138	-	976
ISCFPS1100 U	68.98	114.9	1 100	25 297	146	-	977
ISCFPS1160 U	72.8	121.33	1 160	26 559	155	-	977
ISCFPS1450 U	76.7	127.8	1 450	27 984	163	-	977
ISCFPS1505 U	69.6	116.1	1 505	44 597	148	-	1 109
ISCFPS1593 U	74	123.38	1 593	47 530	157	-	1 110
ISCFPS1648 U	78.3	104.4	1 648	39 304	133	-	1 108
ISCFPS1687 U	78.39	130.65	1 687	50 198	166.45	-	1 110
ISCFPS1775 U	82.76	137.93	1 775	52 851	175	-	1 110
ISCFPS1860 U	87.14	145.23	1 860	55 493	185	-	1 110
ISCFPS 2250 U	91	121.3	2 250	56 688	154.6	-	1 161
ISCFPS2581 U	87.38	134.4	2 581	72 934	170.7	-	1 237
ISCFPS2680 U	88.3	147.2	2 680	70 264	187.4	-	1 125
ISCFPS 2815 U	80.2	133.6	2 815	67 419	170.3	-	1 021
ISCFPS 100 F	36.2	120.6	104	455	153.7	-	461

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(Page 8, Table 5) — Substitute the following table for the existing:

Table 5 Dimensional Tolerances
(Clause 6.1)

a) Hot rolled sections

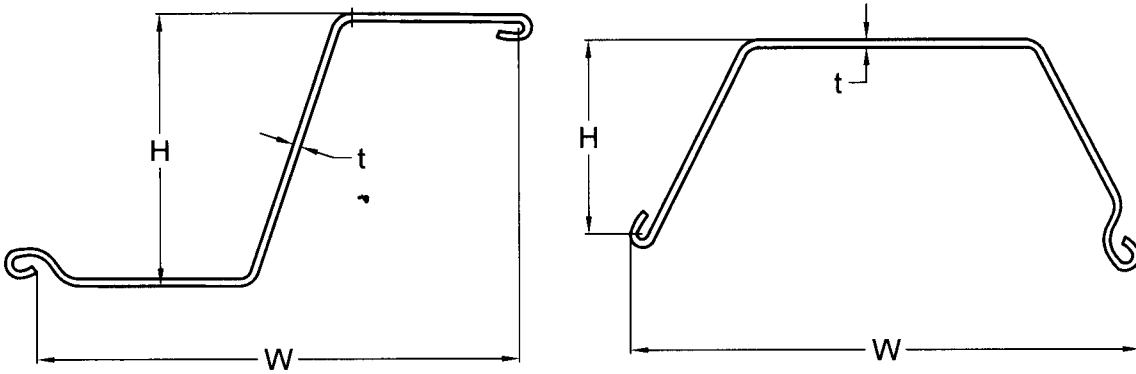


Description Tolerances for (mm)	Z-Type	U-Type	FLAT-Type
Width, W	+ 4.0 - 5.0	+ 4.0 - 4.0	+ 4.0 - 4.0
Height, H	+ 4.0 - 4.0	+ 4.0 - 4.0	- -
Thickness, t t < 10	+ 1.0 - 1.0	+ 1.0 - 1.0	+ 1.5 - 1.0
10 < t < 16	+ 1.2 - 1.2	+ 1.2 - 1.2	+ 1.75 - 1.75
NOTES 1 Width, W in case of Z-type section stands for the appropriate dimension of leg width, C or D given in Table 1. 2 Thickness, t stands for the various thicknesses given in Tables 1, 2 and 3.			

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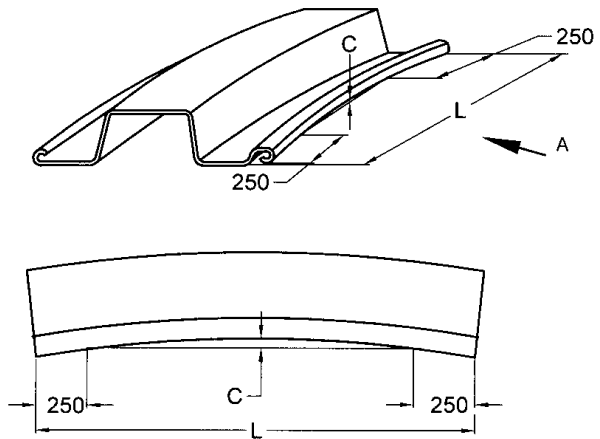
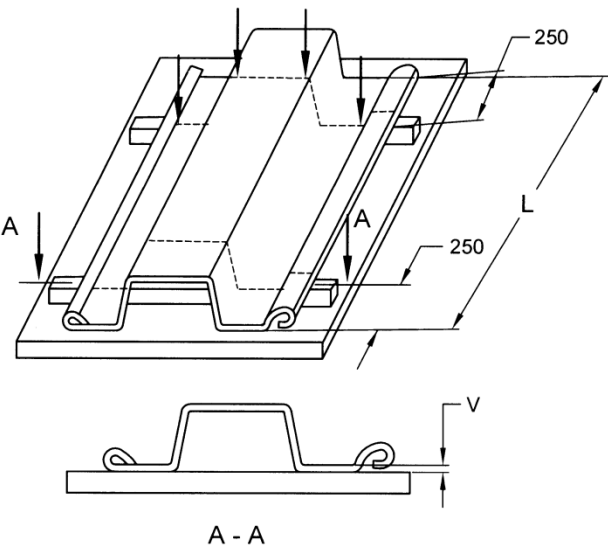
Table 5 Dimensional Tolerances — (Concluded)
(Clause 6.1)

b) Cold formed sections

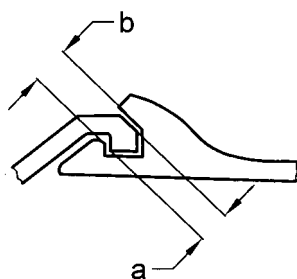


Characteristics	Nominal Size mm	Tolerance mm
Sectional depth, H	$H \leq 200$	± 4
	$200 < H \leq 300$	± 6
	$300 < H \leq 400$	± 8
	$400 < H$	± 10
Sectional width, W	single sheet piles	$\pm 2\% W$
	double sheet piles	$\pm 3\% W$
Sectional thickness, t	$6.00 < t \leq 8.00$	± 0.35
	$8.00 < t \leq 10.00$	± 0.40
Bending Deflection (S)	PLAN VIEW	$\pm 0.25\% L$

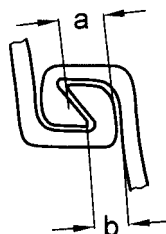
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Curving deflection (C)	 ELEVATION A	$\pm 0.25\% L$
Twist	 A - A	$2\% L$ with 100 mm max
Length		± 50
Squareness of ends [Out-of-squareness (t) of end cuts]		2% of width
Mass of sections (Difference between total actual and total theoretical mass delivered):		$\pm 7\%$

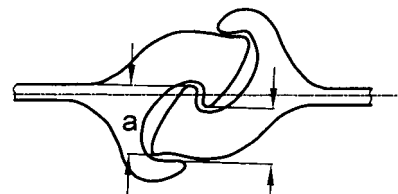
(Page 7, Fig. 4 and Table 6) — Substitute the following for the existing:



a = Button width
b = Interlock opening,
Z - type



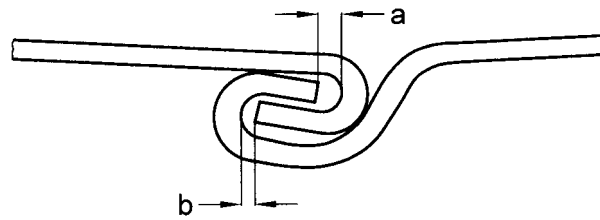
a = Hook width
b = Interlock opening,
U - type



a = Knuckle width
b = Interlock opening,
Flat - type

4A INTERLOCK DIMENSIONS, a AND b (HOT ROLLED)

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4B INTERLOCK DIMENSIONS, A AND B (COLD FORMED)

FIG. 4 INTERLOCK DIMENSIONS, *a* AND *b*

Table 6 Tolerances on Design Dimensions
(Clause 6.2.2)

Type	Design Dimension	Tolerance on Design Dimension	
		mm	mm
(1)	(2)	(3)	(4)
Hot rolled Z-type	Button width, <i>a</i>	+1.0	-3.0
	Interlock opening, <i>b</i>	+3.0	-1.0
Hot rolled U-type	Hook width, <i>a</i>	+2.5	-2.5
	Interlock opening, <i>b</i>	+2.0	-2.0
Hot rolled flat-type	Knuckle width, <i>a</i>	+2.0	-3.0
	Interlock opening, <i>b</i>	+3.0	-2.0
Cold formed Z-type, U-type and flat-type	Hook width, <i>a</i>	+1.5	-1.5
	Interlock opening, <i>b</i>	+1.5	-1.5